Hong Vinh Vu

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About

Control and Automation Engineering undergraduate at HUST with research experience in intelligent control (Fuzzy Logic, Neural Networks) and dynamic optimization (MPC, Reinforcement Learning). Driven to specialize in Reinforcement Learning and optimal control for complex autonomous systems and robotics.

Education

- **Hanoi University of Science and Technology** — Hanoi, Vietnam
  B.Sc. in Control and Automation Engineering; GPA: 3.45 / 4.0 — *Sep. 2024 – Present*
- **Le Quy Don High School for the Gifted** — Vietnam
  Specialized Physics Program. — *Aug 2021 – May 2024*
  Won the Second Prize, Provincial Physics Olympiad.

Experiences

- **Research Assistant** — Hanoi, Vietnam
  Multi-Agent System Control (MASC) Research Laboratory, SEEE, HUST. — *Mar 2026 – Present*
  - **Supervisors**: Assoc.Prof. Hoai Nam Nguyen
  - **Research topic**: Applied Optimal Control Algorithms and Intelligent Control Algorithms to optimize trajectory planning for Quadrotors.
  - **Key Contribution**: Developed a hybrid flight control framework by integrating PyTorch-based Deep Reinforcement Learning (DRL) for high-level dynamic trajectory planning with Model Predictive Control (MPC) and ANFIS for optimal trajectory tracking and nonlinear disturbance compensation.
- **Research Assistant** — Hanoi, Vietnam
  Mechatronics Engineering Group (MEG), — *Jul 2025 – Mar 2026*
  Motion Control and Applied Robotics Laboratory (MoCAR LAB), SEEE, HUST.
  - **Supervisors**: Assoc.Prof. Danh Huy Nguyen
  - **Research topic**: Kinematic Modeling and Control of a 6WD-6WS Mobile Robot based on ROS2.
  - **Key Contribution**: Developed mathematical kinematic/dynamic models and motion control algorithms for a 6WD-6WS (6-Wheel Drive, 6-Wheel Steering) omnidirectional vehicle, implementing custom ROS2 navigation nodes, Gazebo physics simulations, and embedded MCU firmware for real-time trajectory tracking and multi-mode steering coordination.

Projects

- **PID Controller for Quadrotors**: Designed and tuned a discrete PID controller in MATLAB/Simulink for 6-DOF Quadrotor position and attitude stabilization.
- **Fuzzy-PD Controller for Quadrotors**: Developed a Self-Tuning Fuzzy-PD controller that dynamically adjusts K_p and K_d gains based on real-time tracking errors, significantly improving dynamic response and overshoot reduction.
- **Direct Adaptive Fuzzy Controller for Quadrotors**: Implemented a Direct Adaptive Fuzzy Control (DAFC) scheme using Lyapunov stability theory to approximate unmodeled dynamics and external disturbances, achieving robust trajectory tracking with minimal steady-state error.

Honors & Awards

- **Academic Achievement Scholarship** — Apr 2025
  Grade: Excellent. — *Semester 2024.2*
  This is a prestigious award given to students with the highest scores in the entire department ($\approx 3\%$).

Skills

- **Languages**: Python, C, C++, MATLAB/Simulink.
- **Optimization**: PyTorch.
- **Tools**: LaTeX, STM32CubeIDE.
- **Robotics**: ROS2 frameworks, Gazebo, SLAM.